UNITED ARAB EMIRATES
MINISTRY OF EDUCATION
الإمارات العربية المتحدة
وزارة التربية والتعليم

Al Khalidiya Boys School – Cycle 3

School code 10048 | Dibba Al Hisn | Sharjah Eastern

Grade 10 Physics Final Exam Practice

Term 1 2026–2027 • Set B • With Answers

100 marks | 150 minutes | Calculator allowed

Answer all 20 questions. Choose one option for each MCQ. Show working, units and labelled diagrams for Q17–20. Use g = 9.8 m/s² when needed.

Section A Multiple Choice 60 marks

1. A ray makes an angle of 30° with a mirror surface. Its angle of reflection is: [4 marks]

A. 30°
B. 90°
C. 60°
D. 15°

Reference: pp. 4–5; p. 8, Check Your Progress 9,11 | **Answer C**

2. Which is a property of a plane-mirror image? [4 marks]

A. Always enlarged
B. Real and inverted
C. Located on the mirror surface
D. Virtual and the same size as the object

Reference: pp. 5–8 | **Answer D**

3. An object is between F and C of a concave mirror. The image is: [4 marks]

A. Beyond C, real, inverted and enlarged
B. At C and upright
C. Behind the mirror and diminished
D. Between F and the mirror, virtual and smaller

Reference: pp. 11–13 | **Answer A**

4. A concave mirror has radius 16.0 cm and object distance 20.0 cm. Find X_i. [4 marks]

A. +8.0 cm
B. +13.3 cm
C. −13.3 cm
D. +80.0 cm

Reference: p. 16, Practice Problem 14 | **Answer B**

5. Which magnification represents an upright enlarged image? [4 marks]

A. +0.50
B. −0.50
C. +2.0
D. −2.0

Reference: p. 18 | **Answer C**

6. The refractive index is n = c/v. Which implication follows for an ordinary transparent material with n > 1? [4 marks]

A. Light travels more slowly than in vacuum
B. Light travels faster than in vacuum
C. Light stops at the boundary
D. Light has zero frequency

Reference: p. 22 | **Answer A**

7. At the critical angle, the refracted ray is: [4 marks]

A. Along the incident ray
B. Along the boundary
C. Normal to the boundary
D. Inside the denser medium at 45°

Reference: pp. 23–25 | **Answer B**

8. In ordinary glass, which visible colour is usually deviated most by a prism? [2 marks]

A. Violet
B. Red
C. All colours equally
D. Only green

Reference: pp. 24–25 | **Answer A**

9. A hot-road mirage can make sky light appear to come from the road because: [4 marks]

A. Liquid water must be present on the road
B. Refraction through hot air bends the light path toward the observer
C. The light frequency becomes zero
D. The road emits a perfect picture of the sky

Reference: pp. 23–25 | **Answer B**

10. In a ray diagram for a concave lens, a ray initially parallel to the axis emerges: [4 marks]

A. Back along its incident path
B. Through the far focus
C. Along the axis in all cases
D. As if it came from the near focus

Reference: pp. 26,28 | **Answer D**

11. How can chromatic aberration be reduced? [4 marks]

A. Increase dispersion with a single prism
B. Replace all lenses by rough glass
C. Use an achromatic combination of lenses
D. Remove the aperture completely

Reference: p. 31 | **Answer C**

12. Which lens corrects hyperopia? [4 marks]

A. An opaque lens
B. Convex/converging
C. A plane mirror
D. Concave/diverging

Reference: pp. 32–33; p. 35, Check Your Progress 57 | **Answer B**

13. At a dark interference fringe, the two equal-amplitude waves arrive: [4 marks]

A. With different speeds in the same air
B. In phase
C. With zero frequency
D. Out of phase by 180°

Reference: pp. 42–44 | **Answer D**

14. Light reflecting from a boundary into a medium with a higher refractive index undergoes: [4 marks]

A. A change to zero frequency
B. A doubling of wavelength in the incident medium
C. No possible phase change
D. A 180° phase change

Reference: pp. 46–47 | **Answer D**

15. A reflection grating consists of: [4 marks]

A. Many closely spaced reflecting lines or grooves
B. A single broad slit only
C. One isolated mirror without markings
D. A prism with no repeated structure

Reference: pp. 53–54 | **Answer A**

16. How can a spectrum help identify a gemstone sample? [2 marks]

A. Only the brightness matters
B. Every gemstone must have the same spectrum
C. Its spectral features can be compared with known samples
D. A spectrum gives the exact mass without calibration

Reference: pp. 53–55 | **Answer C**

Set B Section B Free Response

17. A light ray strikes a plane mirror at 38° to the normal. Its angle of incidence increases by 13°. Also consider the convex mirror below. [10 marks]

Reference: p. 6, Practice Problem 4; pp. 9–14

(a) Calculate the original and new angles of reflection. [2]

Answer Original 38° [1]; new 51° to the normal [1].

(b) At the new incidence angle, find the angle between the incident ray and the mirror surface. Explain your calculation. [2]

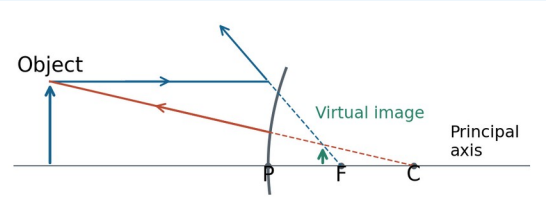
Answer 90° − 51° [1] = 39° [1].

(c) Define focal length and relate it to radius of curvature for a spherical mirror. [2]

Answer Focal length is distance from pole to principal focus [1]; |f| = |R|/2 [1].

(d) Complete a labelled two-ray convex-mirror diagram; state two image properties and one use. [4]

Answer Correct reflected rays with dashed backward extensions [1]; image located behind mirror between P and F [1]; upright and diminished (also virtual) [1]; vehicle/security mirror for wide field of view [1].



18. A ray passes from air into water (n = 1.33) and refracts at 25° to the normal. A separate converging lens of focal length 5.0 cm has a 2.0 cm object 25 cm in front of it. Use X_o and X_i. [10 marks]

Reference: p. 21, Practice Problem 31; p. 30, Practice Problem 46

(a) Find the angle of incidence in air. [3]

Answer 1.00 sin θ_1 = 1.33 sin25° [1]; sin θ_1 = 0.562 [1]; θ_1 = 34.2° [1].

(b) Find the critical angle for water to air. [2]

Answer sin θ_c = 1/1.33 [1]; θ_c = 48.8° [1].

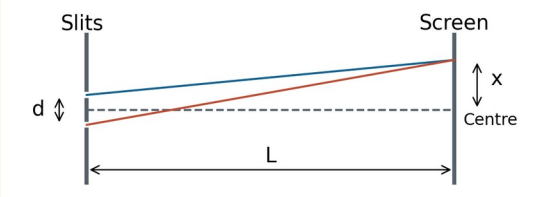
(c) Calculate the lens image distance. [3]

Answer 1/f = 1/X_o + 1/X_i [1]; X_i = 5(25)/(25 − 5) [1] = +6.25 cm [1].

(d) Calculate magnification and image height; identify image orientation. [2]

Answer m = −6.25/25 = −0.25 [1]; h_i = −0.50 cm, inverted [1].

19. In a double-slit investigation the slit separation is 0.0190 mm, the screen is 0.600 m away, and the first bright band is 21.1 mm from the central band. [10 marks]



Reference: p. 45, Example Problem 1; pp. 42–44; p. 49, Check Your Progress 13

(a) On the sketch, identify d, L and x, and state the condition for a first-order bright band. [3]

Answer d: slit separation; L: screen distance; x: distance from central band [2 for all three, 1 for two]. Path difference = λ [1].

(b) Calculate the wavelength using the small-angle approximation. [3]

Answer λ = xd/L [1]; convert x and d to metres [1]; λ = 668 nm [1].

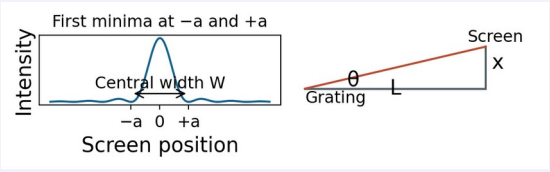
(c) Find the distance of the second-order bright band from the central band. [2]

Answer x_2 = 2x_1 [1] = 42.2 mm [1].

(d) Describe the central fringe and nearby fringes if white light replaces the single wavelength. [2]

Answer Central fringe is white [1]; nearby fringes are coloured, with violet closer to centre than red [1].

20. A He–Ne laser (λ = 632.8 nm) gives a central single-slit bright band 15.0 mm wide on a screen 1.45 m away. Separately, light of wavelength 632 nm passes through a grating. Its first-order bright band is 5.6 cm from the centre on a screen 0.55 m away. [10 marks]



Reference: p. 53, Practice Problem 17; p. 56, Practice Problem 24

(a) Calculate the single-slit width. [3]

Answer w = 2λL/W [1]; w = 2(632.8 × 10^{-9})(1.45)/0.0150 [1] = 1.22 × 10^{-4} m = 0.122 mm [1].

(b) State where the central bright band ends. [1]

Answer At the first minimum on each side of the centre [1].

(c) Find the grating's first-order diffraction angle using exact geometry. [2]

Answer θ = $\tan^{-1}$(0.056/0.55) [1] = 5.81° [1].

(d) Calculate the grating spacing and number of slits per centimetre. [4]

Answer d = λ/sin θ [1] = 6.24 × 10^{-6} m [1]; N = 0.010/d [1] ≈ 1.60 × 10^3 slits/cm [1].